Glaze Search · which model for which input

Version 2026-11-06 (v1) | colour match 70cd01c0c673 | smart search 78afd7dfbe2c | 395 held-out recipes · 6,625 searched | app: Auto engine, `configs/glaze_decision.json`

A decision record: what the updated `test_similarity (3).zip` contains, whether anything in it should be trained again or added to the app, how its model compares with smart search and with two hybrids on recipes that no model trained on, and which engine the app should use for each kind of input. Every number below comes from `python scripts/eval_glaze_engines.py` (133 s on the CPU).

The decision in short

- **No new model in test_similarity (3).** Its 3 model files are byte-identical to (2) and to the installed model (fingerprint 70cd01c0c673); `latent_space.py` is a report on that same model. Nothing needs training again.
- **Photo, words or formula given → Smart search**, by a wide margin on every measure that finds the recipe (e.g. typed sentence R@10 0.79 vs colour match 0.43).
- **Hybrid (smart + the colour-match learned match): not adopted.** Its best weight was 0 and 0.05 on the two halves: adding the colour-match match does not help smart search.
- **Colour match + a new image head: the colour-match space *can* take a cut-out photo** (photo R@10 0.19 → 0.27), but it stays below smart search (0.45). Experiment only.
- **Only a colour → colour ranking without a learned model** (CIEDE2000 + cone + surface). There both learned models lose: MRR 0.136 vs smart 0.113 and colour match 0.079; same colour in the top 10 0.88 vs 0.59.
- **Colour + menus only → Colour match**: With only the colour, cone and surface menus, colour match gives the most results with both the same colour and the same surface (63 % vs 54 % smart search); it finds the exact recipe a little less often (MRR 0.31 vs 0.34).
- **The app now has an Auto engine** (default on card G) that applies this tree; Smart search and Colour match stay as buttons.

1 · What is in test_similarity (3), and does anything need training or applying?

File	What it is	Changed since (2)?	Train again?	In the app? Where
`shared_space_heads.pt`, `glaze_retrieval_vectors.npz`, `glaze_retrieval_catalog.csv`	the trained text head (MiniLM 384 → 256), chemistry head (PCA-16 → 256) with its PCA, the recipe vectors and catalog	no: byte-identical	no	yes: Colour match, installed by `INSTALL_GLAZE_MODEL.bat` to `multimodal_experiments/models/glaze_retrieval`
`retrieval.ipynb`	the search alone (`retrieve_glazes`, Best matching / Explore with `diversify`)	new file	no (no training in it)	yes: `mm/glaze_retrieval.py`, same results as the notebook
`training.ipynb`	the training notebook (masked InfoNCE, composition groups, K-Means by silhouette)	renamed only (was `glaze_kmeans_color_weighted_full_catalog.ipynb`), byte-identical; its training cell is not run in the file	no: the shipped heads are its output	its method: smart search training (`scripts/train_glaze_joint.py`)
`glaze_chemistry_models.joblib`	PCA + scaler + K-Means	saved again: only the last digits of the K-Means centres differ; the cluster labels are the same	no	not loaded: the app reads the clusters from the catalog and the PCA from the .pt
`training_data.csv`	the training table	no	no	no (training input)
`latent_space.py` + `latent_space_out/`	a report on the shared space: alignment, text → own chemistry R@k, modality gap, t-SNE	not in the zip: added to the folder after unpacking	no: it trains nothing	no; its numbers are in section 4 of this document
`__MACOSX/`	macOS zip metadata	–	–	ignore

So the "higher result" of `latent_space.py` is the same model measured again: text → own chemistry R@10 7.0 % on 1,309 test recipes without masking same-group recipes, next to 8.2 % in the Evaluation PDF (D4) with same-group recipes left out. Same model, slightly stricter count.

2 · The decision tree, in the order of the activity diagram

green = the pretrained model or engine the app uses; the green deepens from the input to the best engine at the bottom. grey = tried or measured, not chosen (with the reason or its number).

Start
What do you have? A photo, words, a formula, a colour?

▼ fork: each input goes to its own lane

A photo

A1 · find the objects
Grounding DINO base
zero-shot · prompted with ceramic words

not chosen: the earlier box pipeline (A1–A3 replaced together: 148 of 168 reviewer-flagged photos better, 2 worse)

A2 · cut them out
SAM 2.1 large
zero-shot · box prompts

A3 · is it ceramic?
OpenCLIP ViT-B/32
zero-shot · 30 sentences, keep p ≥ 0.35

A4 · colours
no model
ColorThief + CIEDE2000 names

A5 · six attributes
Grounding DINO evidence → logistic regression
frozen + trained · form 83.6 % (guess 55.5 %)

not chosen: v1 heads on frozen CLIP + DINOv2 features (replaced by A5)

not chosen: fine-tuned Grounding DINO decoder (experiment only: 1,504 labelled objects)

photo → recipe space
DINOv2 ViT-L/14 + trained head
frozen + trained · photo + colour: R@10 0.45

not chosen: DINOv2 → new head into the colour-match space: R@10 0.27

Words

B1 · clean the recipe tags
no model
text-branch rules (cones, spelling)

B3 · read typed words
no model
rules + typo fixing · cone 99.3 %, colour 93.3 %, surface 89.1 % right

B2 · words → recipe space
all-MiniLM-L6-v2 + trained head
frozen + trained · typed sentence → recipe R@10 74.2 %

not chosen: MiniLM alone, untrained: 62.0 %

not chosen: colour-match text head (text → own chemistry R@10 8.2 %)

A formula

oxides → recipe space
no pretrained model: 26 oxides → √ fractions → trained head
trained · formula → recipe R@10 33.1 %

not chosen: PCA-16 → colour-match chemistry head (no formula box for it in the app)

▼ join: everything you gave goes to the recipe search

Which recipe engine? (Auto decides by what you gave, decision of 2026-11-06)

a photo, words or a formula

BEST FOR THIS INPUT · USED BY AUTO
Smart search
Colour + words (clean): MRR 0.55 · Colour + typed sentence (typos): MRR 0.53 · Photo object + colour: MRR 0.22 · Photo + typed sentence + colour: MRR 0.56
With a photo, words or a formula, smart search finds the recipe far more often (typed sentence: in the top 10 for 79 % vs 43 % with colour match) and gives more results of the same colour and surface.

Hybrid · MRR 0.54 · MRR 0.51 · MRR 0.21 · MRR 0.56

Colour ranking · MRR 0.34 · MRR 0.27 · MRR 0.14 · MRR 0.27

Colour match · MRR 0.31 · MRR 0.24 · MRR 0.08 · MRR 0.24

only the colour + cone / surface menus

BEST FOR THIS INPUT · USED BY AUTO
Colour match (team model)
Colour + cone / surface menus: colour+surface 0.63
With only the colour, cone and surface menus, colour match gives the most results with both the same colour and the same surface (63 % vs 54 % smart search); it finds the exact recipe a little less often (MRR 0.31 vs 0.34).

Smart search · colour+surface 0.54

Hybrid · colour+surface 0.54

Colour ranking · colour+surface 0.55

only a colour

BEST FOR THIS INPUT · USED BY AUTO
Colour ranking (no learned model)
Colour only: same colour 0.88
With only a colour, the plain CIEDE2000 colour ranking beats both learned models: 88 % of the results in the same colour (smart search 59 %) and the recipe found more often (MRR 0.14 vs 0.11).

Smart search · same colour 0.59

Hybrid · same colour 0.60

Colour match · same colour 0.87

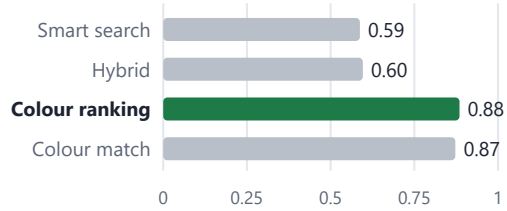
Pots instead of recipes? A6 · Similar pottery page: DINOv2 look + silhouette shape + surface attributes + colour regions (ΔE2000), weights tuned on validation queries.

3 · Evaluation: every engine on recipes no model trained on

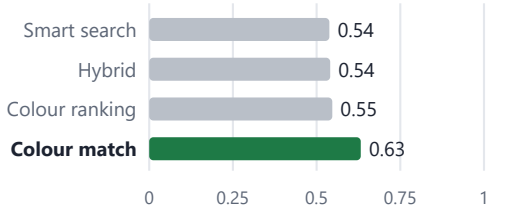
Set-up. Test recipes: in the colour-match *test* split **and** in the joint model's *validation or test* split, with a measured colour: **395** recipes. Every recipe of both catalogs can be found (6,625). Each test recipe is asked six ways (below). Its colour is moved by a random CIELAB offset (sd 3 per channel, median ΔE00 3.7), as a new photo of the same glaze would be. In the photo queries the query object is removed from the recipe's own gallery embedding (leave-one-object-out; rebuilt embeddings match the installed ones, cosine 1.000000). The two hybrid weights are cross-fitted: chosen on one half of the test recipes (split by composition group), scored on the other half.
Measures. R@1 / R@10: the true recipe at #1 / in the top 10 (random R@10 = 0.0015). MRR: mean of 1 / rank. Same chemistry: a recipe in the top 10 whose square-root oxide fractions have cosine ≥ 0.99 with the true one. Same colour: ΔE00 ≤ 10 to the true colour. Colour AND surface: both at once. Differences with a paired bootstrap (2,000 resamples), 95 % interval.

■ the engine Auto uses for this input ■ other engines · each panel shows the measure that decides that input: what the input asks for (only a colour → the same colour; colour + menus → the same colour and surface; a photo, words or formula → the right recipe)

Colour only · same colour in the top 10



Colour + cone / surface menus · same colour AND surface in the top 10



Colour + words (clean) · MRR: 1 / rank of the right recipe



Colour + typed sentence (typos) · MRR: 1 / rank of the right recipe

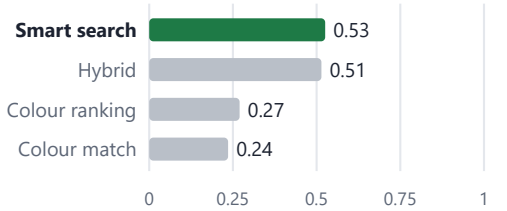


Photo object + colour · MRR: 1 / rank of the right recipe

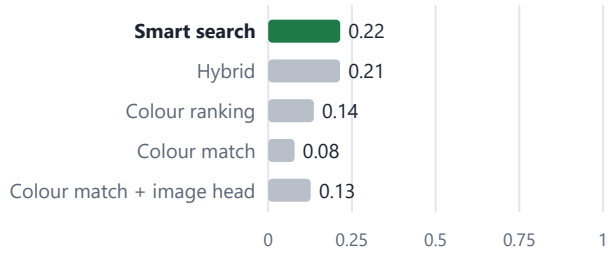
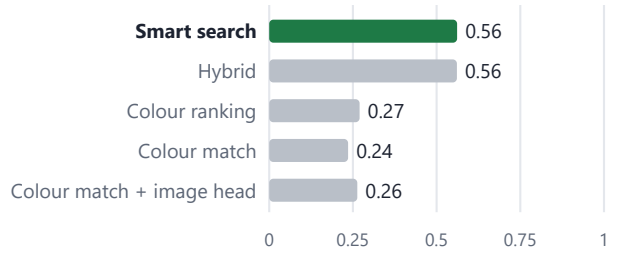


Photo + typed sentence + colour · MRR: 1 / rank of the right recipe



All measures

Colour only · n = 395	R@1	R@10	MRR	same chemistry in top 10	same colour	same surface	colour AND surface
Smart search	0.048	0.238	0.113	0.324	0.588	**0.210**	0.122
Hybrid (smart + colour-match learned match)	0.041	0.240	0.105	0.322	0.597	0.209	0.123
Colour ranking (no learned model)	**0.063**	**0.279**	**0.136**	**0.370**	**0.885**	0.186	**0.161**
Colour match (team model)	0.028	0.187	0.079	0.268	0.873	0.180	0.152

Colour + cone / surface menus · n = 395	R@1	R@10	MRR	same chemistry in top 10	same colour	same surface	colour AND surface
Smart search	0.218	**0.600**	**0.343**	0.658	0.569	**0.966**	0.538
Hybrid (smart + colour-match learned match)	0.215	**0.600**	0.343	**0.663**	0.572	**0.966**	0.541
Colour ranking (no learned model)	**0.225**	0.559	0.343	0.638	**0.862**	0.658	0.547
Colour match (team model)	0.208	0.547	0.309	0.635	0.795	0.854	**0.632**

Colour + words (clean) · n = 395	R@1	R@10	MRR	same chemistry in top 10	same colour	same surface	colour AND surface
Smart search	**0.398**	**0.820**	**0.550**	**0.863**	0.820	**0.954**	**0.724**
Hybrid (smart + colour-match learned match)	0.387	0.810	0.545	0.858	0.818	0.954	0.723
Colour ranking (no learned model)	0.225	0.559	0.343	0.638	**0.862**	0.658	0.547
Colour match (team model)	0.203	0.549	0.309	0.630	0.800	0.850	0.634

Colour + typed sentence (typos) · n = 395	R@1	R@10	MRR	same chemistry in top 10	same colour	same surface	colour AND surface
Smart search	**0.392**	0.792	**0.526**	0.841	0.788	**0.788**	0.578
Hybrid (smart + colour-match learned match)	0.370	**0.795**	0.514	**0.846**	0.789	0.788	**0.579**
Colour ranking (no learned model)	0.162	0.478	0.270	0.557	**0.865**	0.566	0.477
Colour match (team model)	0.147	0.430	0.235	0.514	0.813	0.695	0.535

Photo object + colour · n = 395	R@1	R@10	MRR	same chemistry in top 10	same colour	same surface	colour AND surface
Smart search	**0.104**	0.448	**0.215**	0.625	0.774	0.290	0.224
Hybrid (smart + colour-match learned match)	**0.104**	**0.453**	0.214	**0.630**	0.779	**0.291**	**0.225**
Colour ranking (no learned model)	0.063	0.279	0.136	0.370	0.885	0.186	0.161
Colour match (team model)	0.028	0.187	0.079	0.268	0.873	0.180	0.152
Colour match + new image head	0.063	0.268	0.127	0.367	**0.892**	0.192	0.169

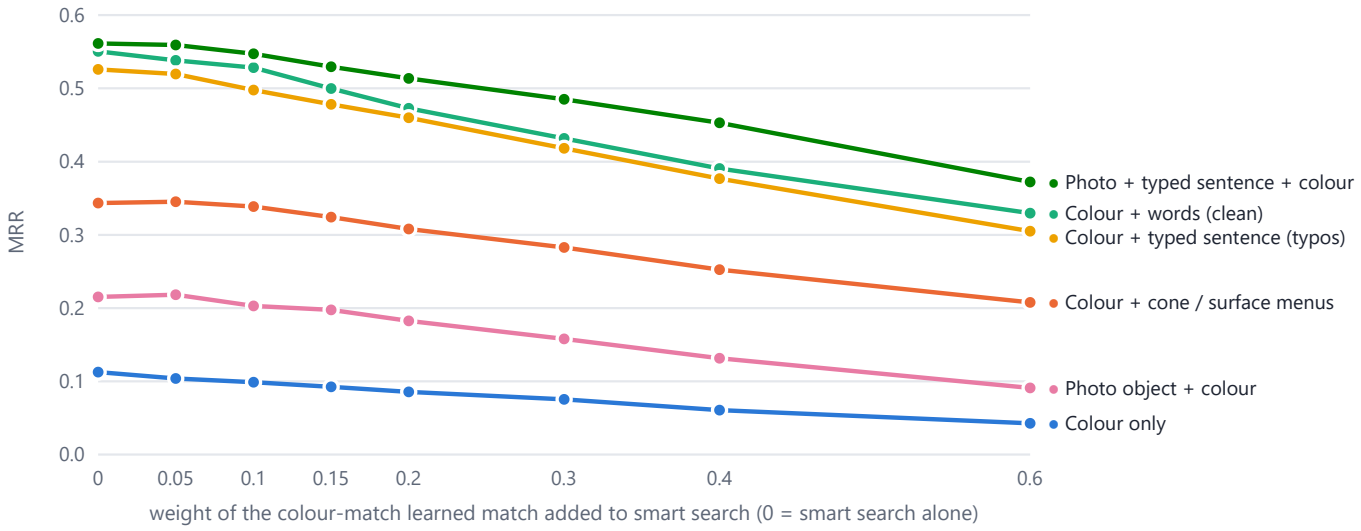
Photo + typed sentence + colour · n = 395	R@1	R@10	MRR	same chemistry in top 10	same colour	same surface	colour AND surface
Smart search	**0.428**	**0.838**	**0.561**	**0.906**	0.795	0.776	0.584
Hybrid (smart + colour-match learned match)	**0.428**	0.835	0.560	0.904	0.796	**0.776**	**0.585**
Colour ranking (no learned model)	0.162	0.478	0.270	0.557	**0.865**	0.566	0.477
Colour match (team model)	0.147	0.430	0.235	0.514	0.813	0.695	0.535
Colour match + new image head	0.172	0.483	0.263	0.570	0.817	0.694	0.532

Is the difference real? (paired bootstrap, 95 % interval)

Way of asking	Comparison	MRR	R@10	same colour	colour AND surface
Colour only	rules vs smart	+0.024 [+0.009, +0.038]	+0.041 [+0.010, +0.073]	+0.297 [+0.278, +0.317]	+0.040 [+0.028, +0.051]
Colour only	rules vs colour	+0.058 [+0.038, +0.077]	+0.091 [+0.053, +0.129]	+0.012 [-0.001, +0.025]	+0.009 [-0.007, +0.024]
Colour + cone / surface menus	rules vs smart	-0.001 [-0.026, +0.026]	-0.041 [-0.076, -0.003]	+0.293 [+0.276, +0.311]	+0.009 [-0.019, +0.036]
Colour + cone / surface menus	colour vs smart	-0.035 [-0.068, -0.001]	-0.053 [-0.101, -0.008]	+0.227 [+0.207, +0.247]	+0.094 [+0.068, +0.119]
Colour + words (clean)	smart vs colour	+0.242 [+0.203, +0.278]	+0.271 [+0.223, +0.317]	+0.019 [+0.003, +0.038]	+0.091 [+0.071, +0.113]
Colour + typed sentence (typos)	smart vs colour	+0.290 [+0.250, +0.329]	+0.362 [+0.311, +0.413]	-0.025 [-0.045, -0.004]	+0.043 [+0.020, +0.065]
Photo object + colour	smart vs colour	+0.137 [+0.108, +0.168]	+0.261 [+0.208, +0.317]	-0.099 [-0.121, -0.077]	+0.071 [+0.050, +0.093]
Photo object + colour	colour+img vs colour	+0.048 [+0.031, +0.066]	+0.081 [+0.046, +0.119]	+0.019 [+0.006, +0.033]	+0.017 [+0.003, +0.029]
Photo object + colour	colour+img vs smart	-0.088 [-0.118, -0.059]	-0.180 [-0.235, -0.129]	+0.118 [+0.098, +0.139]	-0.055 [-0.074, -0.035]
Photo + typed sentence + colour	smart vs colour	+0.326 [+0.282, +0.367]	+0.408 [+0.359, +0.456]	-0.018 [-0.037, +0.002]	+0.049 [+0.028, +0.069]
Photo + typed sentence + colour	hybrid vs smart	-0.001 [-0.006, +0.005]	-0.003 [-0.013, +0.005]	+0.001 [-0.002, +0.003]	+0.001 [-0.001, +0.004]
Colour + typed sentence (typos)	hybrid vs smart	-0.011 [-0.021, -0.003]	+0.003 [-0.005, +0.010]	+0.001 [-0.003, +0.003]	+0.001 [-0.003, +0.004]

An interval that does not include 0 is a real difference on these 395 recipes.

Hybrid: adding the colour-match learned match to smart search



Every line falls as the colour-match part gets more weight: the learned match adds no information that smart search does not already have. Chosen weight on each half: 0 and 0.05 (best on all test recipes: 0).

The colour-match space with a cut-out photo (new image head)

One linear layer (DINOv2 1,024 → 256, like the colour-match heads) trained on the colour-match *train* recipes only (4,145 objects) against the frozen colour-match chemistry vectors, masked InfoNCE (temperature 0.07), early stop on 281 held-out recipes: photo → own chemistry R@10 **0.379** (random 0.036) after 31 epochs (1.6 s). The team's model is not changed. Added to colour match with weight 0.4 (cross-fitted): photo R@10 0.19 → 0.27, photo + words 0.43 → 0.48. Smart search: 0.45 and 0.84. So the answer to "can the text–chemistry space take cropped images?" is **yes**, and a photo even aligns with chemistry better than the template text does, but the combined engine is still well below smart search. The head is kept with the evaluation (`work/eval/glaze_engines_2026-11-06.image_head.pt`), not installed.

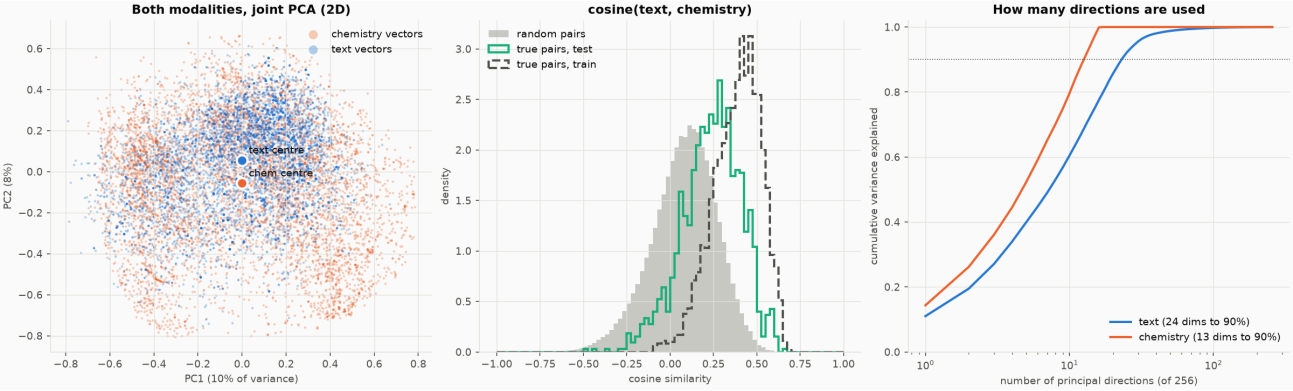
Robustness: does the decision hold?

For each way of asking: the engine Auto uses vs the best other engine, under no colour noise, the main noise, double noise, and on the recipes whose colour is the same in both catalogs (the query colour comes from the joint model's palette, so there no engine has a home advantage). The first row of each input is the measure that decides it. Green = the chosen engine is best or within 0.005 (tie); orange = another engine is better on that measure. The hybrid is left out here: with its cross-fitted weight (0–0.05) it is smart search itself. Two kinds of orange are expected: the menus MRR (the trade-off stated in the decision) and the **no colour noise** column, where the query colour is exactly the colour stored for the recipe, so an engine that leans on colour finds the recipe from its colour alone. No new photo gives the exact stored colour: that column is a stress test, not the main result.

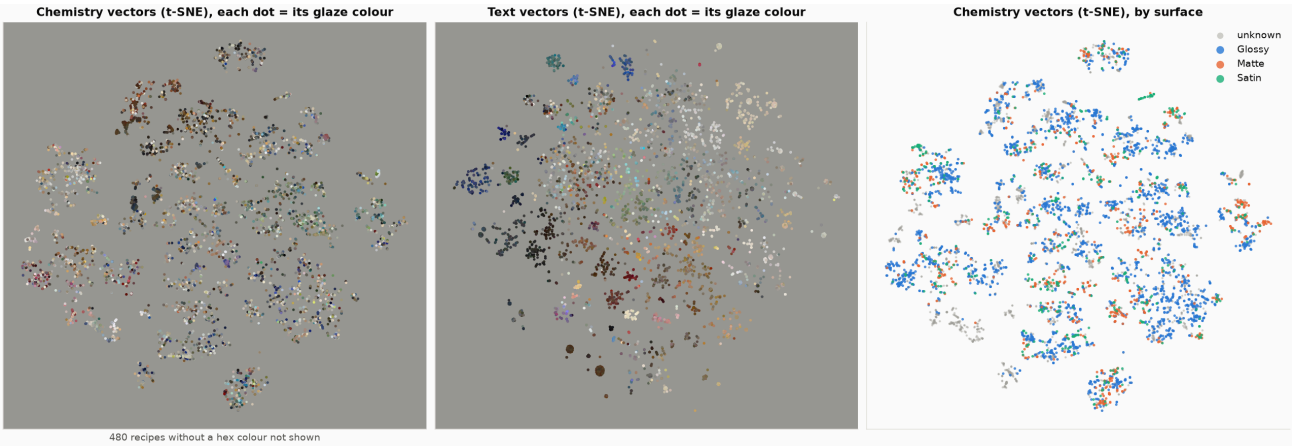
Way of asking	Measure	no colour noise	noise sd 3 (main)	noise sd 6	same colour in both catalogs (327)
Colour only	same colour · **Colour ranking** vs the best other	0.908 vs 0.908 (tie) Colour match	0.885 vs 0.873 Colour match	0.671 vs 0.673 (tie) Colour match	0.889 vs 0.879 Colour match
Colour only	MRR · **Colour ranking** vs the best other	0.825 vs 0.977 Smart search	0.136 vs 0.113 Smart search	0.038 vs 0.035 (tie) Smart search	0.164 vs 0.113 Smart search
Colour + cone / surface menus	colour AND surface · **Colour match** vs the best other	0.689 vs 0.583 Smart search	0.632 vs 0.547 Colour ranking	0.479 vs 0.411 Smart search	0.657 vs 0.556 Colour ranking
Colour + cone / surface menus	MRR · **Colour match** vs the best other	0.698 vs 0.985 Smart search	0.309 vs 0.343 Smart search	0.149 vs 0.151 (tie) Smart search	0.365 vs 0.407 Colour ranking
Colour + words (clean)	MRR · **Smart search** vs the best other	0.958 vs 0.833 Colour ranking	0.550 vs 0.343 Colour ranking	0.303 vs 0.158 Colour match	0.564 vs 0.407 Colour ranking
Colour + words (clean)	colour AND surface · **Smart search** vs the best other	0.758 vs 0.689 Colour match	0.724 vs 0.634 Colour match	0.613 vs 0.499 Colour match	0.728 vs 0.653 Colour match
Colour + typed sentence (typos)	MRR · **Smart search** vs the best other	0.920 vs 0.816 Colour ranking	0.526 vs 0.270 Colour ranking	0.294 vs 0.115 Colour match	0.534 vs 0.321 Colour ranking
Colour + typed sentence (typos)	colour AND surface · **Smart search** vs the best other	0.605 vs 0.573 Colour match	0.578 vs 0.535 Colour match	0.488 vs 0.421 Colour match	0.580 vs 0.548 Colour match
Photo object + colour	MRR · **Smart search** vs the best other	0.740 vs 0.825 Colour ranking	0.215 vs 0.136 Colour ranking	0.074 vs 0.048 Colour match + image head	0.210 vs 0.164 Colour ranking
Photo object + colour	colour AND surface · **Smart search** vs the best other	0.275 vs 0.225 Colour ranking	0.224 vs 0.169 Colour match + image head	0.167 vs 0.138 Colour match + image head	0.223 vs 0.175 Colour match + image head
Photo + typed sentence + colour	MRR · **Smart search** vs the best other	0.870 vs 0.816 Colour ranking	0.561 vs 0.270 Colour ranking	0.375 vs 0.130 Colour match + image head	0.561 vs 0.321 Colour ranking
Photo + typed sentence + colour	colour AND surface · **Smart search** vs the best other	0.607 vs 0.583 Colour match + image head	0.584 vs 0.535 Colour match	0.511 vs 0.421 Colour match	0.581 vs 0.548 Colour match

4 · The latent-space report (latent_space.py), in context

Number	latent_space.py (folder of (3))	Evaluation PDF (D4)	Meaning
text → own chemistry R@10, test	7.0 % (1,309; same-group recipes not masked)	8.2 % (same-group excluded)	same model, slightly stricter count; random 0.8 %
cosine, right pair vs random pair	0.238 vs 0.079 (test)	0.238 vs 0.076	about 3× apart: learned, but weak
effective dimensions	text 20.1 · chemistry 12.5 of 256	–	the chemistry side passes PCA-16, so it can use at most 16 directions
modality gap	centres cosine 0.515	–	text and chemistry vectors sit in two separate clouds (figure below, left)



latent_space_overview.png (from the folder, unchanged). Left: text (blue) and chemistry (orange) vectors in one PCA: two clouds whose centres do not meet. Middle: the right text–chemistry pairs (green, test) score higher than random pairs (grey), but the two overlap a lot. Right: the chemistry side uses about 13 directions, the text side 24, of 256.



latent_space_tsne.png (from the folder, unchanged). Each dot is a recipe coloured by its glaze colour. The text vectors group by colour (the colour word is in the text); the chemistry vectors do not: blues, browns and whites mix inside the same chemistry groups, and surfaces mix too. So the chemistry side alone cannot tell colours apart, and the colour part of the score (ΔE00) has to do that work.

The template text is short and repeats: 3,878 different sentences for 6,718 recipes; 3,662 recipes share their exact sentence with another recipe. No model can tell those recipes apart from the text, which caps the learned match. In section 3 this shows: at finding the right recipe, colour match ranks below the same ranking with its learned part switched off (colour ranking minus colour match, MRR: colour only +0.058, menus +0.034, words +0.034, typed +0.035). What the learned part does add: results a little closer in chemistry (colour only: 0.898 vs 0.890) and, because the template text names the surface, many more results with the asked surface. That is why it gives the best colour-and-surface list when only the menus are used (0.63 vs 0.55).

5 · What changed in the app, and how the page talks to the server

Part	Before	Now (2026-11-06)
Card G · 4 How to search	Smart search (default) · Colour match	**Auto (best for your input)** (default) · Smart search · Colour match
Page → server	`POST /api/glaze/smart_search` or `POST /api/glaze/search`	Auto: `POST /api/glaze/auto` with the smart-search body (text, colour, cone, surface, photo object, formula). The other two buttons are unchanged.
Server	–	`glaze_auto()` in `app/server.py`: `mm/glaze_decision.py` reads what was given → smart search, colour match, or colour ranking (colour match with weights {"latent": 0.0, "color": 0.8, "surface": 0.1, "cone": 0.1})
Server → page	results, parts, versions	the same, plus `engine` (smart / colour) and `decision` {version, input, engine, why}; the page shows "Auto (decision of 2026-11-06): … → …" above the results
`GET /api/glaze/options`	both models' settings	plus `decision` (the whole decision file): the engine line shows the rules and their date
Search log	mode "smart search" / "colour match"	Auto searches are logged as "auto 2026-11-06: <engine>"
Smart search code	scoring inside `search()`	`JointSearch.score()` returns every recipe's score parts (used by the evaluation); results byte-identical, checked on 6 queries

A new evaluation: run `python scripts/eval_glaze_engines.py`, write the new date and engines into `configs/glaze_decision.json`, rebuild this document under the new date. No code change is needed for a new decision.

6 · Versions

Version	Date	What

v1	2026-11-06	First decision: test_similarity (3) checked file by file; colour match, smart search, colour ranking, two hybrids evaluated on 395 held-out recipes, six ways of asking, three colour-noise levels; Auto engine added to card G.

Generated from `multimodal_experiments/work/eval/glaze_engines_2026-11-06*.json` (copy next to this file: `CG2_Glaze_Search_Model_Decision_2026-11-06.json`). Seed 0.